

for electronic information using one address type to be delivered to devices using the other.

Commenting on the beginning of the IPv6 era, Prem Nithin, business development manager, Borderless Networks, Cisco India, said, "If you look at things on the ground, the era of IPv4 is not yet over for enterprises. Most of the enterprise customers are seriously looking to have IPv6 deployed at their organisation but there is no compelling reason, yet, for them to do so. There is no deadline-linked need to have IPv6 as of today, because these enterprises are still content with IPv4. Having said that, I will not deny that they are looking for testing IPv6 majorly, creating their applications accordingly and also incorporating it in their parallel set-up.

"Verticals like the BFSI and IT/ITES, which are large consumers of IT equipment, are seriously exploring IPv6 as an option. The BFSI sector, in particular, is doing that because of the mandate from the RBI. Till last year, they were only *talking* about IPv6 but RBI guidelines have now forced them to be IPv6 compliant by the end of March. So this sector is seriously looking at what are the dependencies when it comes to enabling IPv6 on the Internet, and what they will have to do on the application side, the networking side, for firewalls, routers and all the other things involving IPv6. We are aggressively involved with financial institutions in terms of helping them enable IPv6. Now what they do for their internal networks is something that has still not been taken up very seriously by them because their primary focus is to get their Internet apps working on IPv6. Their payment gateways, and Visa and Mastercard gateways have to be IPv6 ready. They will only turn their focus onto the internal aspects of the transition after they get the Internet edge on IPv4. They would then want to come out with a strategy as to what to do on the internal network. They will have two choices: they can either do a dual stack of IPv4 and IPv6 or they can get away with IPv4. These discussions have just started."

IT managers are one of the major target audiences for companies working on offering products that help transition from IPv4 to IPv6. K P Unnikrishnan, APAC marketing director of Brocade Communications, says, "Transition from IPv4 to IPv6 is not something that can happen at one go. It has to be in stages, and IT managers have to be well prepared for it before they venture into it. Before an IT manager plans to take his company onto IPv6, he has to understand the technical aspects involved in this transition. IT managers are well aware of the fact that they have to be educated on both levels—at the technical level as well as the business level."

He adds, "As end users, IT managers will come across several product and service providers offering IPv6 solutions. So you need to be well informed technically, academically and on the business front—enough to select

a solution. The deployment has to be done phase-wise. So, one cannot really make a judgement on the investment required for this job. Nobody does a 100 per cent shift, end-to-end. With an investment of US\$ 50,000 to 100,000 you can start implementing IPv6 deployment in a phased manner in your company."

Things you must consider when migrating to IPv6

It is practically not possible, or desirable, to replace all your existing computing and networking equipment with IPv6 equipment at once. But what is expected from IT managers is that when they buy new IT equipment during regular technology upgrades or while the company is scaling up, they can ensure that the new equipment is IPv6-ready.

Migrating to IPv6 is as much of a business decision as a technical one. So when you decide on when to and how to migrate to IPv6, you should conduct both a business analysis and a technical analysis. While you are doing the technical evaluation, it is important to understand the options being offered and also evaluate which ones suit your business better. We have listed some of these technical options below.

ISP-provided IPv6 network address translation services

If you do not want to convert any of your existing IPv4 routers to IPv6-compliant ones right away, you can still communicate with your customers and suppliers. Unnikrishnan emphasises, "Firms can rely on ISPs that offer network address translation (NAT) services to keep them connected to IPv6 devices outside their own network. NAT can be thought of as a special forwarding service that can read one type of IP address and figure out what the corresponding address is for the other type. Since the applications that many companies rely on to run their businesses are using IPv4 in the data centre and campus (LAN), there is no pressing need to migrate, as long as they can continue to connect to their customers and suppliers over the Internet via an ISP-supplied NAT service."

Dual-stack routers

Dual-stack routers come to the rescue of all IT managers looking to get IPv6-ready without a major investment. All they need to ensure is that the new network routers they purchase are capable of operating with both IPv4 and IPv6 address types. This is called 'dual stack'. Unnikrishnan says, "In this case, networking protocols are layered in a stack like a cake. At the routing layer, the equipment can handle IPv4 or IPv6 address types and traffic; hence, they are dual-stack routers. This does

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